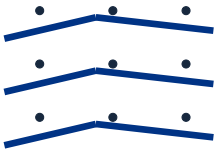
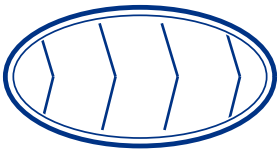
Organelle structure and composition

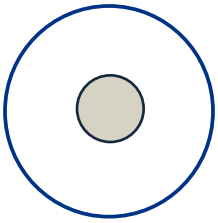
Biomolecules build cell structures; chemistry helps explain what each part does.



Rough ER + ribosomes



Mitochondrion



Nucleolus

Component	Built from	Structure -> function
Membrane bilayer	Phospholipids + proteins; other lipids vary	Water-facing heads and a hydrophobic core form a selective boundary.
Ribosome	rRNA + proteins	Non-membranous translation machine; rRNA contributes to catalysis.
Chromatin	DNA + associated proteins	Stores genetic information and helps package/regulate it.
Cytoskeleton	Protein filaments	Supports shape, movement and intracellular organization.
Cristae	Folded inner mitochondrial membrane	Provides membrane area for respiratory-chain complexes and ATP synthase.

Monomer, polymer and lipid are different ideas

Protein: amino acids linked by peptide bonds. DNA/RNA: nucleotides linked by phosphodiester bonds. A typical glycerophospholipid has glycerol, two fatty-acid tails and a phosphate-containing head; it is not a repeating-chain polymer.

Organelle structure and composition

Self-check questions and explanations. Refer to the visual guide on the front.

How are rough ER, mitochondria, nucleolus and a bilayer distinguished?

Rough ER has membrane sacs with ribosomes; mitochondria have outer and folded inner membranes; the nucleolus is a non-membranous nuclear region; a bilayer has two phospholipid leaflets.

Which mitochondrial feature supports oxidative phosphorylation?

Cristae are folds of the inner mitochondrial membrane, which contains respiratory-chain components and ATP synthase. More area can accommodate more of this machinery.

What materials build a ribosome and a bilayer?

Ribosomes contain rRNA and proteins. Phospholipids have water-interacting heads and hydrophobic tails; a bilayer shields tails from water while heads face the aqueous sides.

What are protein and nucleic-acid monomers and bonds? Is a phospholipid a polymer?

Proteins: amino acids, peptide bonds. Nucleic acids: nucleotides, phosphodiester bonds. A typical glycerophospholipid: glycerol, two fatty-acid tails and a phosphate-containing head. It is not a repeating monomer-chain polymer.

How does the nucleolus help form ribosomes?

rRNA is made and processed, and ribosomal subunits assemble with proteins in the nucleolus. rRNA has structural/catalytic roles; ribosomal proteins support ribosome structure and function. Subunits are exported.

How can a membrane protein span both water and the lipid core?

A transmembrane protein can have hydrophobic side chains in the lipid core and hydrophilic regions facing fluid. Its chemical regions match the surrounding membrane or aqueous environment.